



Arc fit

Turn freeform coplanar edges into a chain of lines and circular arcs within a position tolerance -- for CAM and for radius callouts.

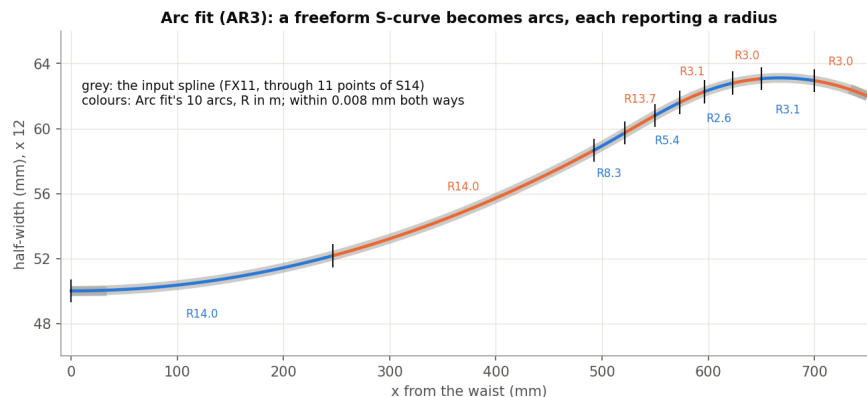


What it does

Machines and drawings want lines and arcs; design curves are splines. Arc fit replaces each edge with the fewest lines and arcs that stay within tolerance, keeping edges that already are arcs or lines.

Use it when

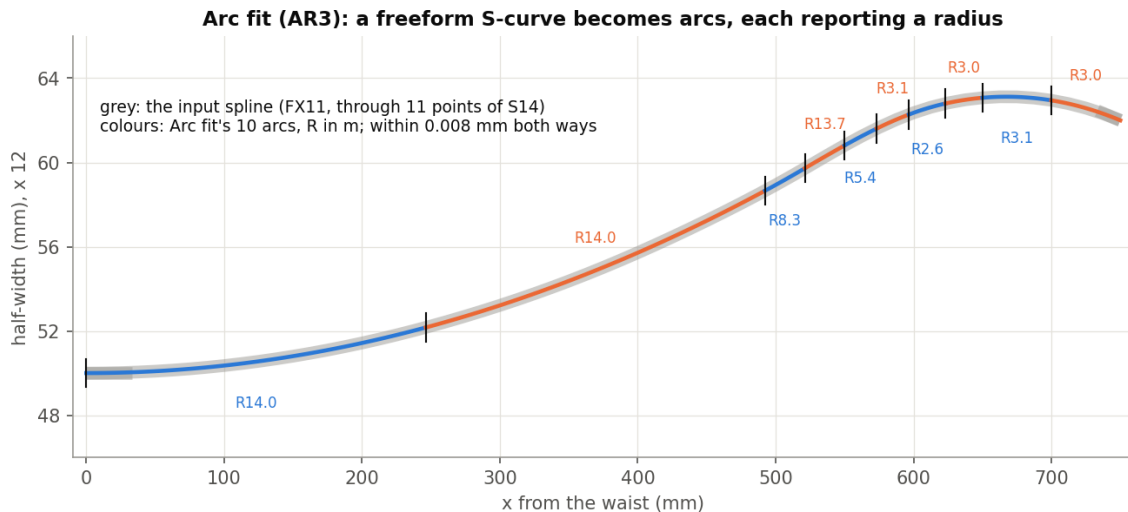
- A spline sidecut needs radius callouts
- CAM needs line/arc geometry
- Check how 'arc-like' a curve is



AR3: a spline S-curve -> 10 arcs within 0.008 mm.



How it fits



Keep what is exact

An edge that already is a circle or line passes through whole.

Biarcs

Each segment: a line, a biarc (two tangent arcs matching the end tangents) or a 3-point arc.

Split, then merge

Split at the worst point until in tolerance; merge neighbours while it stays in tolerance.



The dialog

IMAGE MISSING:
docs/decks/arc_fit/shots/dialog.png

#	Parameter	What it means
1	Edges to fit	Coplanar edges, selected in chain order.
2	Position tolerance	Largest deviation (0.01 mm); also the join tolerance.
3	Plane tolerance	Allowed out-of-plane error (0.01 mm).
4	Minimum segment length	1 mm.
5	Tangency tol	No effect today.
6	Validation samples per arc	Where the deviation is checked (16).
7	Max subdivision depth	8.
8	Output type	Curves / Sketch / Curves and sketch.
9	Debug	Console diagnostics.



Examples (Footprint tests)

AR1: 5 exact arcs

Passed through: R 0.3 / 3 / 14 / 3 / 0.3 m.

AR2: spline on an R14 arc

One arc R14000.

AR4: tiny knot spans

2 mm spans at the inflection, minimum 10 mm: still one wire, no gaps.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The arc-fit wire (empty when Output type is Sketch).

Messages

	Message / what to do
Error	edges not coplanar (deviation stated) Raise Plane tolerance or fix the input.



Tips and limits



Order matters

Select the edges in chain order; they are not re-ordered.



Across edges

Arcs are tangent within an input edge, not across input edges.



Tight tolerances

Very small tolerances drive every segment to maximum depth: slow.



Where it lives

Document	footprint (52724f3a857fa52d3ecceb77)
Tab	arcFit
Tests	Footprint tests studio (build_ / check_footprint_tests.py): AR1-AR4
Explainer	docs/explainers/footprint/footprint_explained.md, section 2.4
Related	Integrate footprint, Analyze footprint, Scale Footprint, Arc fit, Generate Footprint Points